

Predicting Antihistamine Activity Using Machine Learning: A Comparative Study of Structural and Physicochemical Descriptors

ITC3133 - Data Mining and Big Data
final project

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Table of contents:

Abstract.....	3
Literature Review.....	3
Materials and Methods.....	5
Data.....	5
Pre-processing and Data Preparation.....	6
Modelling Strategy.....	7
Parameter Tuning.....	7
Validation and Evaluation.....	8
Statistical Comparison.....	8
Results and Discussion.....	9
Model Performance on the Merged Dataset.....	9
Comparison Across Datasets.....	11
Validation Comparison: 10-Fold Cross-Validation vs LOOCV.....	13
ROC Analysis.....	14
Statistical Significance Analysis.....	16
Overall Interpretation.....	17
Conclusion and future work.....	17
References.....	19
Appendix.....	20
Appendix A: Statistical Significance Analysis.....	20
Appendix B: WEKA Knowledge Flow for Random Forest classification and evaluation.....	23

Abstract

Predicting the biological activity of chemical compounds is a key challenge in drug discovery, where machine learning methods can reduce the need for costly experimental screening. This study evaluates the effectiveness of machine learning models in predicting antihistamine activity using molecular descriptors. Two types of features were considered: structural fingerprints and physicochemical properties, which were analyzed individually and as a combined dataset to assess their impact on classification performance.

Three classification algorithms were employed: J48 decision tree, Random Forest, and Logistic Regression. Data preprocessing and dataset merging were conducted using Python, while modelling, validation, and evaluation were performed in WEKA. Model performance was primarily assessed using 10-fold cross-validation, with Leave-One-Out Cross-Validation (LOOCV) used for additional robustness analysis. Evaluation metrics included accuracy, F-measure, mean absolute error, and Cohen's kappa. Statistical significance testing was conducted using a corrected paired t-test in the WEKA Experimenter.

The results show that Random Forest achieved the best overall performance, reaching approximately 81% accuracy on the merged dataset, while Logistic Regression performed competitively after tuning. J48 showed lower and less stable performance. The merged dataset generally produced more consistent results, indicating that combining descriptor types improves predictive capability.

These findings demonstrate the effectiveness of ensemble methods and highlight the importance of feature integration in improving model performance.

Literature Review

Machine learning has become a fundamental approach in modern drug discovery, particularly in the prediction of the biological activity of chemical compounds. The increasing availability of chemical and biological data has enabled the application of data-driven techniques to accelerate the identification of promising drug candidates. In this context, Quantitative Structure–Activity Relationship (QSAR) modelling plays a central role, as it establishes relationships between molecular descriptors and biological activity, allowing compounds to be classified as active or inactive (Dara et al., 2022; Roy, 2017). This approach significantly reduces reliance on costly and time-consuming experimental screening by leveraging existing datasets to generate predictive models. Recent advances in machine learning have further enhanced the ability to process complex and high-dimensional chemical data, improving predictive accuracy and efficiency in preclinical drug discovery (Catacutan et al., 2024).

A wide range of machine learning algorithms has been applied to drug activity prediction, among which Random Forest (RF) and Support Vector Machines (SVM) are the most widely studied. Random Forest is particularly effective due to its ensemble nature, combining multiple decision trees to model nonlinear relationships while reducing overfitting (Dara et al., 2022). This characteristic makes it highly suitable for chemical datasets, which are often noisy and heterogeneous. In contrast, SVMs are well-suited to high-dimensional feature spaces and are capable of constructing optimal decision boundaries between classes. Comparative studies indicate that while both models perform well, Random Forest tends to provide more robust performance when handling mixed and noisy molecular data, whereas SVM may perform better in structured datasets with clear class separation (Catacutan et al., 2024). These findings highlight the importance of selecting models that align with the characteristics of the dataset.

In addition to model selection, the choice of molecular descriptors is a critical factor influencing predictive performance. Molecular descriptors are typically categorized into structural and physicochemical features. Structural descriptors, often represented as binary fingerprints, indicate the presence or absence of specific molecular substructures. These descriptors are widely used due to their efficiency in capturing structural similarities between compounds and their suitability for large-scale screening (Roy, 2017). Physicochemical descriptors, on the other hand, are continuous numerical variables that describe properties such as molecular size, polarity, hydrophobicity, and electronic distribution. These features provide quantitative insight into molecular behavior and are essential for understanding interactions within biological systems (Lange et al., 2024).

Previous QSAR studies have examined the effectiveness of these descriptor types individually. Structural fingerprints are particularly effective in capturing structural patterns and are well suited for classification tasks; however, they may lack detailed quantitative information. Conversely, physicochemical descriptors provide deeper insight into chemical properties but may not fully represent structural similarities among compounds (Evangelista & Papa, 2025). As a result, relying on a single descriptor type may limit the overall predictive performance of machine learning models.

Recent research trends have increasingly focused on integrating multiple descriptor types to improve model performance. The rationale behind this approach is that structural and physicochemical descriptors provide complementary information, and their integration offers a more comprehensive representation of molecular characteristics. Studies in cheminformatics have demonstrated that models trained on combined feature sets often achieve higher accuracy and improved generalization compared to those using a single descriptor type (Lange et al., 2024; Dara et al., 2022). Despite these promising findings, integrated approaches are not consistently applied across all domains, and their effectiveness may vary depending on the dataset and problem context.

A key limitation in existing literature is the lack of standardized comparisons between descriptor types and modelling approaches under identical experimental conditions. Many studies evaluate structural and physicochemical descriptors independently, making it difficult to determine the true benefit of combining them. Furthermore, relatively limited research has focused specifically on predicting antihistamine activity, highlighting a gap in the application of machine learning techniques to this class of compounds (Catacutan et al., 2024).

Therefore, although previous studies have demonstrated the effectiveness of machine learning models and molecular descriptors in predicting drug activity, there remains a need to investigate the combined impact of structural and physicochemical features specifically for antihistamine classification. This study aims to address this gap by integrating both descriptor types into a unified dataset and evaluating their influence on model performance. By comparing multiple machine learning models across different dataset representations, the study provides a comprehensive assessment of how feature integration affects predictive accuracy and robustness.

Despite the advances in machine learning for QSAR modelling, predicting antihistamine activity remains a challenging problem due to the complexity of molecular structures and the variability of descriptor representations. Existing studies often rely on either structural fingerprints or physicochemical properties independently, making it unclear whether combining these descriptor types can improve predictive performance. Therefore, the key problem addressed in this study is to determine how different molecular representations and machine learning models influence the accuracy and robustness of antihistamine activity prediction.

Materials and Methods

Data

This study investigated the prediction of antihistamine activity using two provided datasets and one derived merged dataset. The classification task is formulated as a supervised binary problem, where each molecule is labeled as either **active** or **inactive**. The datasets consist of molecular descriptors that capture different aspects of chemical structure and behavior.

Two distinct datasets were used. **Dataset 1** contains structural fingerprints represented as binary features, where each feature indicates the presence or absence of a specific molecular substructure. These fingerprint representations are commonly used in cheminformatics because they efficiently encode structural similarity across compounds. **Dataset 2** contains physicochemical properties represented as continuous numerical variables. These features describe measurable molecular characteristics such as molecular size, polarity, hydrophobicity,

and electronic distribution, providing insight into how molecules behave under chemical and biological conditions.

To provide a more comprehensive representation of molecular information, a third dataset was created by merging Dataset 1 and Dataset 2 using a common molecule identifier. This ensured that each instance corresponds to the same compound across both datasets. The merged dataset therefore combines binary structural features with continuous physicochemical descriptors within a unified framework.

The use of all three datasets enables a systematic comparison of feature representations. Dataset 1 allows evaluation of structural information alone, Dataset 2 evaluates physicochemical properties independently, and the merged dataset enables assessment of whether combining these complementary descriptors improves classification performance.

Pre-processing and Data Preparation

Pre-processing was performed using Python, while modelling and evaluation were conducted in WEKA. This workflow was selected because Python provides flexibility for data manipulation, while WEKA offers a structured environment for classification, validation, and statistical analysis. The initial step involved inspecting both datasets to identify the class variable and a common identifier for merging. Duplicate instances were removed to prevent repeated samples from biasing the model. The datasets were then checked for missing values. Where necessary, simple imputation methods were applied to handle missing numeric values, ensuring that no incomplete records remained.

The two datasets were merged using the common molecule identifier. Only compounds present in both datasets were retained to ensure consistency between structural and physicochemical representations. This step ensured that each row in the merged dataset corresponds to a single molecule with both types of descriptors.

Since Dataset 2 contains continuous variables with different numerical scales, normalization was applied to these features. This step is particularly important for algorithms such as Logistic Regression, which are sensitive to feature magnitude. Normalization ensures that all features contribute proportionally to the model and prevents dominance of variables with larger scales.

After merging, additional checks were performed to ensure data integrity. Duplicate columns, including potential duplicate class attributes, were removed to avoid data leakage. This step was critical to ensure that the models do not have direct or indirect access to the target variable during training. The resulting datasets were verified to contain only valid features and a single class label.

Three final datasets were prepared for analysis:

- **Dataset 1:** structural fingerprints only
- **Dataset 2:** physicochemical properties only
- **Merged dataset:** combination of structural and physicochemical features

All datasets were exported in CSV format and imported into WEKA for classification experiments.

Modelling Strategy

Three supervised classification algorithms were selected: **J48**, **Random Forest**, and **Logistic Regression**. These models were chosen to represent different modelling approaches and levels of complexity.

J48 is a decision tree algorithm that constructs a tree structure by recursively splitting the dataset based on feature values. It produces interpretable classification rules, allowing insight into how specific features influence the prediction. As a result, J48 serves as an interpretable baseline model.

Random Forest is an ensemble learning method that builds multiple decision trees using random subsets of both data and features. The final prediction is obtained by aggregating the outputs of all trees. This approach reduces overfitting and improves generalization, making it particularly suitable for high-dimensional datasets with mixed feature types.

Logistic Regression is a linear probabilistic model that estimates the relationship between input features and the probability of class membership. It is particularly effective when relationships between variables and the target class are approximately linear. Logistic Regression also provides insight into feature contributions through model coefficients.

Together, these models enable comparison between interpretable rule-based learning, nonlinear ensemble learning, and linear probabilistic modelling.

Parameter Tuning

Parameter tuning was conducted to optimize model performance. Tuning was initially performed on the merged dataset using 10-fold cross-validation, as this dataset provides the most comprehensive feature representation.

For **J48**, the parameters explored included the confidence factor, which controls pruning, the minimum number of instances per leaf, and whether pruning is applied. These parameters directly affect tree complexity. Lower confidence factors and higher minimum leaf sizes result in simpler trees, while reduced pruning allows more complex trees that may capture more patterns.

For **Random Forest**, the parameters explored included the number of trees (iterations), the number of features considered at each split, and the maximum tree depth. Increasing the number of trees generally improves model stability, while controlling depth can prevent overfitting. The number of features per split influences the level of randomness and diversity among trees.

For **Logistic Regression**, tuning focused on the ridge regularization parameter and the maximum number of iterations. Regularization helps control overfitting, particularly in high-dimensional datasets, by penalizing large coefficient values. The maximum number of iterations ensures convergence during training.

After identifying optimal configurations on the merged dataset, these settings were applied to Dataset 1 and Dataset 2 to allow consistent comparison across different feature representations.

Validation and Evaluation

Model performance was evaluated using **10-fold cross-validation**, which is a widely used validation technique that divides the dataset into ten subsets. In each iteration, nine subsets are used for training and one for testing. This process is repeated until all subsets have been used for testing, providing a robust estimate of model performance.

In addition to 10-fold cross-validation, **Leave-One-Out Cross-Validation (LOOCV)** was applied to the best-performing model. LOOCV uses all but one instance for training in each iteration and tests on the remaining instance. This method provides a near-unbiased estimate of performance and is particularly useful for datasets of limited size.

The evaluation metrics used in this study were:

- **Accuracy**, measuring the proportion of correctly classified instances
- **Mean Absolute Error (MAE)**, representing the average prediction error
- **F-measure**, capturing the balance between precision and recall
- **Cohen's kappa**, measuring agreement between predicted and actual classifications beyond chance

Statistical Comparison

To determine whether differences in model performance were statistically significant, the WEKA Experimenter was used. A **corrected paired t-test** with a significance level of 0.05 was applied.

Comparisons were conducted across all models and datasets using repeated runs. Each model was used as a baseline in turn, allowing pairwise comparisons between classifiers. This approach enables identification of statistically significant differences in performance rather than relying

solely on observed metric values. This statistical analysis provides a framework for evaluating model performance and supports reliable conclusions regarding model effectiveness.

Results and Discussion

Model Performance on the Merged Dataset

The initial phase of experimentation focused on tuning model parameters using the merged dataset, as it combines both structural and physicochemical descriptors and provides the most comprehensive representation of molecular properties.

For **J48**, the best-performing configurations achieved an accuracy of **75.54%**, with an F-measure of **0.75** and kappa value of **0.51**. These results were obtained using both a less-pruned tree (confidence factor = 0.5) and an unpruned tree.

Table 1. Performance of J48 under different parameter configurations on the merged dataset using 10-fold cross-validation.

Iteration	ConfidenceFactor	minNum Obj	unpruned	accuracy	MAE	F-measure	kappa
1	0.25	2	false	73.35	0.28	0.73	0.46
2	0.1	2	false	72.62	0.29	0.72	0.45
3	0.5	2	false	75.54	0.26	0.75	0.51
4	0.25	5	false	74.81	0.28	0.74	0.49
5	0.25	10	false	72.26	0.32	0.72	0.44
6	0.25	2	true	75.54	0.26	0.75	0.51

As shown in Table 1, reducing pruning improves model performance, indicating that a more flexible tree structure better captures patterns in the dataset. However, increasing the minimum number of instances per leaf or applying stronger pruning reduces performance, suggesting that overly simplified trees fail to capture data complexity. Despite these improvements, J48 consistently underperformed compared to the other models, indicating that a single decision tree is insufficient for modelling the relationships in the merged dataset.

For **Random Forest**, performance was consistently higher and more stable across configurations. The baseline model already achieved **80.65% accuracy**, and increasing the

number of trees to 200 improved performance to **81.02%**, with an F-measure of **0.81** and kappa of **0.62**.

Table 2. Performance of Random Forest under different parameter configurations on the merged dataset.

Iteration	numIterations	numFeatures	maxDepth	accuracy	MAE	F-measure	kappa
1	100	0	0	80.65	0.29	0.80	0.61
2	50	0	0	80.65	0.29	0.80	0.61
3	200	0	0	81.02	0.29	0.81	0.62
4	100	10	0	80.29	0.29	0.80	0.60
5	100	50	0	78.83	0.28	0.78	0.57
6	100	0	5	75.54	0.40	0.75	0.51
7	100	0	10	81.02	0.34	0.80	0.62
8	500	10	10	81.02	0.34	0.81	0.62

As shown in Table 2, increasing the number of trees improves performance up to a point, after which accuracy stabilizes, indicating diminishing returns. Limiting tree depth to 5 significantly reduces performance due to underfitting, while increasing the number of features per split excessively reduces the beneficial randomness of the ensemble. The selected configuration (**500 trees, 10 features, depth 10**) maintains strong and stable performance, confirming that Random Forest effectively captures nonlinear relationships while preserving generalization.

For **Logistic Regression**, performance improved significantly after tuning. The default configuration achieved only **70.07% accuracy**, but increasing the ridge parameter improved performance to **80.29% accuracy**, with an F-measure of **0.80** and kappa of **0.60**.

Table 3. Performance of Logistic Regression under varying regularization and iteration settings on the merged dataset.

Iteration	ridge	maxIts	accuracy	MAE	F-measure	kappa
1	1.0e-8	-1	70.07	0.30	0.69	0.40
2	0.01	-1	78.46	0.22	0.78	0.57

3	0.1	-1	78.46	0.22	0.78	0.57
4	1	-1	78.83	0.24	0.78	0.57
5	10	-1	79.93	0.26	0.79	0.59
6	0.1	100	78.46	0.23	0.78	0.57
7	10	100	79.93	0.26	0.79	0.59
8	100	100	80.29	0.32	0.80	0.60
9	1000	100	79.93	0.42	0.79	0.59

As shown in Table 3, increasing regularization improves model stability and performance, highlighting its importance in high-dimensional datasets. Although Logistic Regression performed competitively, it remained slightly below Random Forest, indicating that nonlinear relationships play a significant role in this problem.

Comparison Across Datasets

Applying the tuned models to all three datasets provided insight into the impact of feature representation.

For **J48**, the merged dataset achieved **75.54% accuracy**, compared with **71.53%** on Dataset 1 and **76.64%** on Dataset 2. This suggests that the merged representation does not significantly improve performance for single decision trees.

Table 4. Performance comparison of J48, Random Forest, and Logistic Regression on Dataset 1.

	ConfidenceFactor	minNumObj	unpruned	accuracy	MAE	F-measure	kappa
J48	0.5	2	false	71.53	0.30	0.71	0.43
	0.25	2	true	71.89	0.30	0.72	0.44
	numIterations	numFeatures	maxDepth	accuracy	MAE	F-measure	kappa
Random forest	200	0	0	80.65	0.30	0.80	0.61

	100	0	10	79.92	0.37	0.79	0.59
	500	10	10	81.38	0.37	0.81	0.63
	ridge	maxIts		accuracy	MAE	F-measure	kappa
Logistic regression	100	100		79.93	0.32	0.79	0.59
	10	-1		79.19	0.26	0.79	0.58

As shown in Table 4, Random Forest achieved the highest performance (**81.38% accuracy**), indicating that structural fingerprints are highly informative for classification. Logistic Regression performed competitively, while J48 showed weaker performance.

Table 5. Performance comparison of J48, Random Forest, and Logistic Regression on Dataset 2.

	ConfidenceFactor	minNumObj	unpruned	accuracy	MAE	F-measure	kappa
J48	0.5	2	false	76.64	0.25	0.76	0.53
	0.25	2	true	76.28	0.25	0.76	0.52
	numIterations	numFeatures	maxDepth	accuracy	MAE	F-measure	kappa
Random forest	200	0	0	76.28	0.31	0.76	0.52
	100	0	10	74.45	0.32	0.74	0.49
	500	10	10	77.74	0.31	0.77	0.55
	ridge	maxIts		accuracy	MAE	F-measure	kappa
Logistic regression	100	100		70.80	0.44	0.70	0.41
	10	-1		69.70	0.40	0.69	0.39

Table 5 shows that Dataset 2 produces lower performance across all models, with Random Forest achieving **77.74% accuracy** and Logistic Regression dropping to **70.80%**. This indicates that physicochemical descriptors alone provide weaker predictive power compared to structural features. Overall, Dataset 1 and the merged dataset provided stronger performance, while Dataset 2 alone was the weakest representation.

Validation Comparison: 10-Fold Cross-Validation vs LOOCV

The best-performing model, Random Forest, was further evaluated using both 10-fold cross-validation and Leave-One-Out Cross-Validation (LOOCV).

Table 6. Comparison of Random Forest performance using 10-fold cross-validation and LOOCV on the merged dataset.

	numIterations	numFeatures	maxDepth	accuracy	MAE	F-measure	kappa
10-fold cv	200	0	0	81.02	0.29	0.81	0.62
	100	0	10	81.02	0.34	0.80	0.62
	500	10	10	81.02	0.34	0.81	0.62
LOO cv	200	0	0	82.48	0.29	0.82	0.65
	100	0	10	79.93	0.34	0.79	0.60
	500	10	10	80.29	0.35	0.80	0.60

As shown in Table 6, LOOCV slightly improves performance in some configurations, such as increasing accuracy from **81.02% to 82.48%** for the 200-tree model. This is expected, as LOOCV uses more training data in each iteration. However, this improvement is not consistent. For example, the selected configuration (**500, 10, 10**) shows a slight decrease under LOOCV (**80.29%**), indicating that LOOCV may introduce higher variance.

Tables 7 and 8 show validation comparisons for Dataset 1 and Dataset 2, respectively.

Table 7. Validation comparison for Dataset 1.

	numIterations	numFeatures	maxDepth	accuracy	MAE	F-measure	kappa
10-fold	200	0	0	80.65	0.30	0.80	0.61

cv							
	100	0	10	79.92	0.37	0.79	0.59
	500	10	10	81.38	0.37	0.81	0.63
LOO cv	200	0	0	81.02	0.30	0.81	0.62
	100	0	10	79.19	0.37	0.79	0.58
	500	10	10	80.65	0.37	0.80	0.61

Table 8. Validation comparison for Dataset 2.

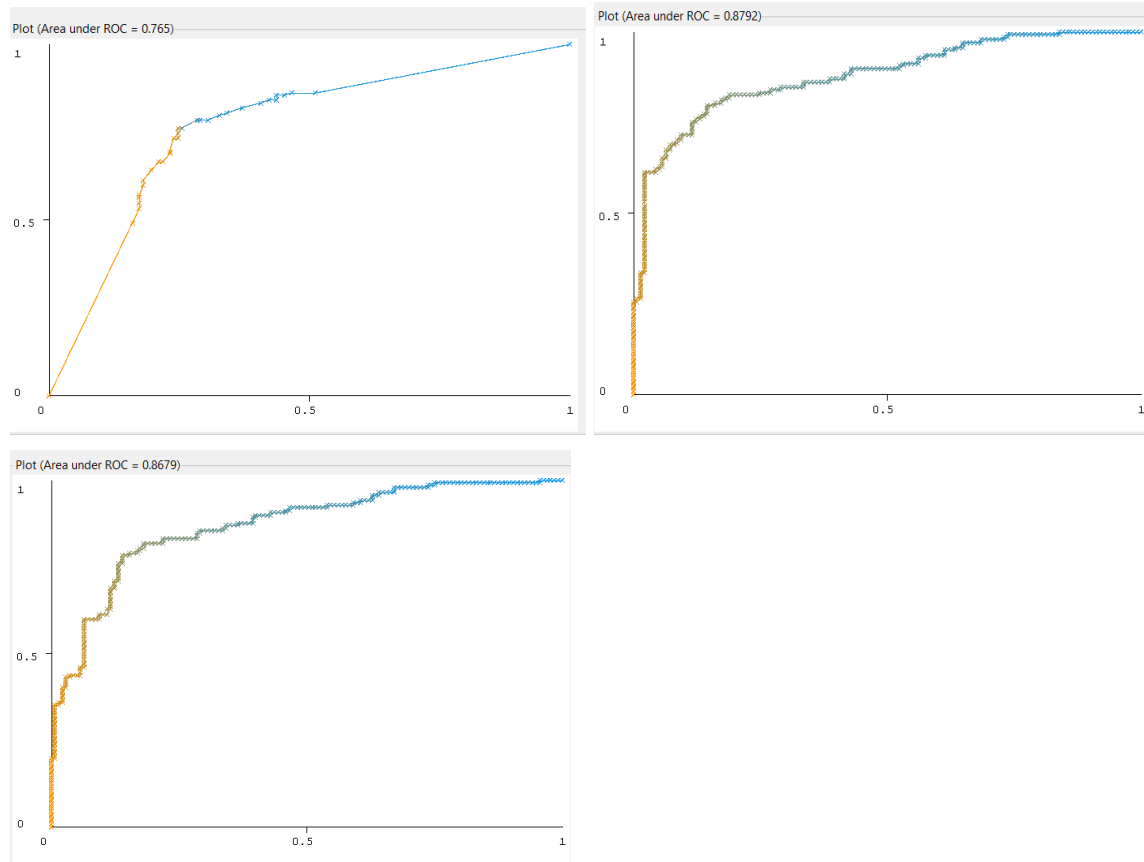
	numIte ra tions	numFeat ures	maxDept h	accurac y	MAE	F-measu re	kappa
10-fold cv	200	0	0	76.28	0.31	0.76	0.52
	100	0	10	74.45	0.32	0.74	0.49
	500	10	10	77.74	0.31	0.77	0.55
LOO cv	200	0	0	77.37	0.31	0.77	0.55
	100	0	10	76.64	0.32	0.76	0.53
	500	10	10	77	0.31	0.77	0.54

These tables show minimal differences between validation methods, suggesting that the model is stable and generalizes well across datasets.

ROC Analysis

To further evaluate model performance, Receiver Operating Characteristic (ROC) curves were analyzed.

Figure 1. ROC curves comparing J48, Random Forest, and Logistic Regression on the merged dataset.



As illustrated in Figure 1, Random Forest in the top right corner achieves the highest area under the curve of 0.879, indicating superior classification performance. Logistic Regression performs competitively with an area of 0.867, while J48 shows weaker discrimination ability with an area of 0.765.

Figure 2. ROC curves for Random Forest across Dataset 1, Dataset 2, and the merged dataset.

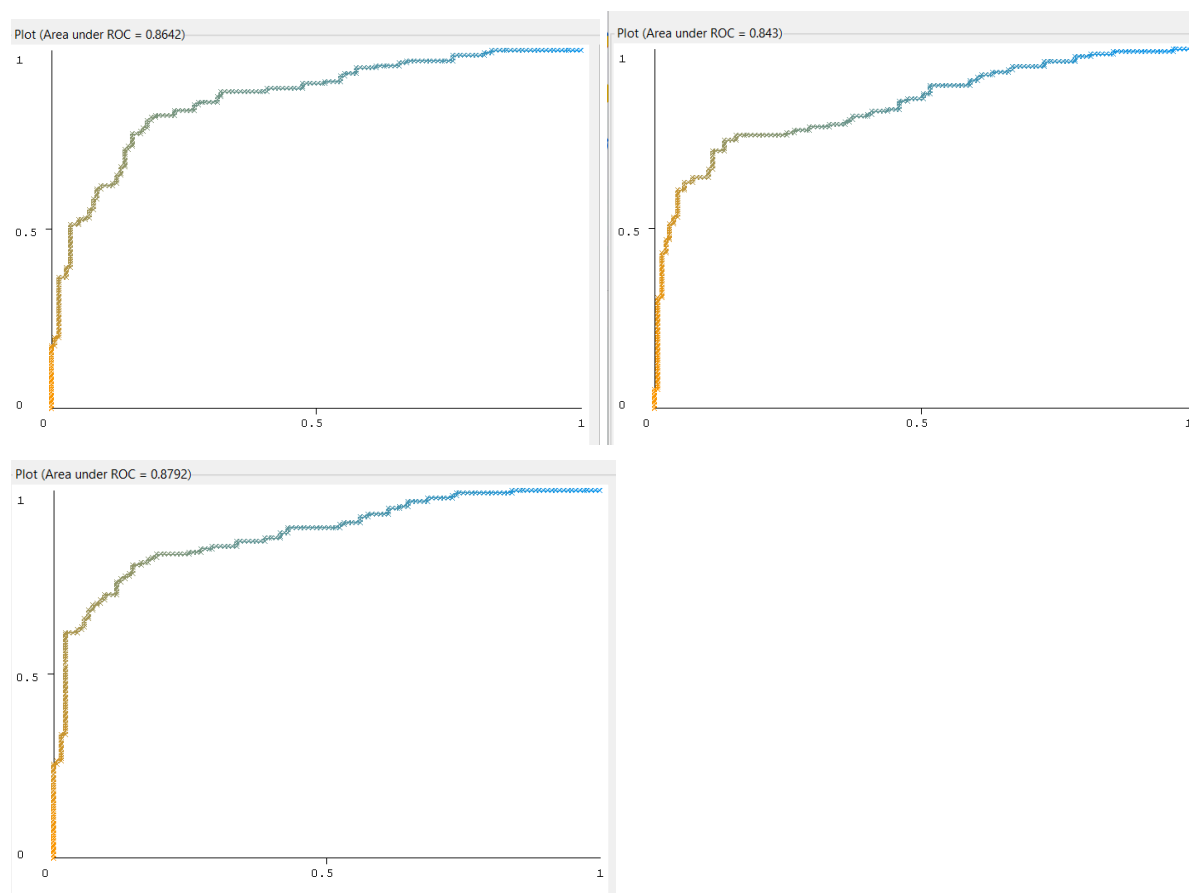


Figure 2 shows that Dataset 1 achieved an area of 0.864 and the merged dataset achieved an area of 0.879, as mentioned before, which are higher ROC values compared to Dataset 2 with an area of 0.843, confirming that structural and combined descriptors provide stronger predictive capability.

Statistical Significance Analysis

Statistical testing using the WEKA Experimenter provided further insights into model comparison. Using a corrected paired t-test at a significance level of 0.05, the analysis showed that performance differences varied across datasets and metrics. When Random Forest was used as the baseline, J48 was significantly worse on the merged dataset, while Logistic Regression was generally not significantly different. In some cases, Logistic Regression outperformed Random Forest on Dataset 2. When J48 was used as the baseline, both Random Forest and Logistic Regression significantly outperformed it on Dataset 1. When Logistic Regression was used as the baseline, Random Forest showed comparable performance overall but performed better on Dataset 2 for certain metrics. These results confirm that Random Forest and Logistic Regression are the most competitive models, while J48 is less stable. The occasional anomalous

results observed for J48 suggest sensitivity to data splits and potential overfitting. All results details are mentioned in the appendix.

Overall Interpretation

The results demonstrate that both model choice and feature representation significantly impact classification performance. Random Forest emerged as the best-performing model due to its ability to capture nonlinear relationships and handle mixed feature types effectively. Logistic Regression also performed strongly after tuning, indicating the presence of linear patterns in the data. In contrast, J48 was limited by its inability to model complex interactions. The dataset comparison shows that combining structural and physicochemical descriptors provides a more comprehensive representation of molecular properties. While Dataset 1 performed strongly, the merged dataset offered consistent performance across models and supported the objective of integrating multiple descriptor types.

Overall, the findings confirm that Random Forest is the most robust model for predicting antihistamine activity and that combining descriptor types enhances performance and generalization.

Conclusion and future work

This study investigated the effectiveness of machine learning models in predicting antihistamine activity using molecular descriptors derived from structural fingerprints and physicochemical properties. The primary objective was to evaluate how different data representations and modelling approaches influence classification performance and to determine whether combining descriptor types improves predictive accuracy and robustness.

The results demonstrate that model choice and feature representation both play a critical role in classification performance. Among the three models evaluated, Random Forest consistently achieved the best overall performance, reaching approximately 81% accuracy on the merged dataset and maintaining strong results across all evaluation metrics, including F-measure and Cohen's kappa. Logistic Regression also performed competitively after appropriate parameter tuning, achieving results close to Random Forest, which indicates that part of the relationship between molecular descriptors and antihistamine activity can be approximated linearly. In contrast, J48 decision trees showed lower accuracy and less stable performance, suggesting that a single tree model is insufficient to capture the complexity of the dataset.

The comparison across datasets provided important insights into the role of descriptor types. Dataset 1, consisting of structural fingerprints, performed strongly, particularly for Random Forest, indicating that structural information is highly informative for predicting antihistamine

activity. Dataset 2, which contains physicochemical properties, produced lower performance across all models, suggesting that these features alone are less predictive. The merged dataset, combining both descriptor types, provided consistently strong and balanced performance across different models, supporting the hypothesis that structural and physicochemical features offer complementary information. Therefore, the project goal of assessing the impact of combined feature representations was successfully achieved, as the results demonstrate that integrating multiple descriptor types enhances model robustness and predictive capability.

Validation experiments further supported these findings. While Leave-One-Out Cross-Validation (LOOCV) occasionally produced slightly higher accuracy than 10-fold cross-validation, the differences were generally small and inconsistent across configurations. This indicates that the selected models are relatively stable and generalize well. As a result, 10-fold cross-validation was considered the more reliable evaluation method due to its balance between bias and variance. Statistical significance testing using the WEKA Experimenter confirmed that Random Forest significantly outperformed J48 in most cases, while its performance was often comparable to Logistic Regression. This reinforces the conclusion that ensemble methods provide a strong and reliable solution for this classification task.

Despite these positive results, several limitations should be acknowledged. First, the dataset size is relatively limited, which may restrict the ability of the models to fully capture complex relationships and may affect the generalizability of the findings. Second, the presence of high-dimensional and sparse binary features in the structural dataset introduces noise, which may impact model stability, particularly for simpler models such as J48. Additionally, some instability was observed in the Experimenter results, where J48 occasionally produced unrealistically high accuracy values, indicating sensitivity to data splits and potential overfitting under repeated validation conditions. These limitations highlight the challenges associated with modelling chemical datasets and suggest areas for improvement.

Future work can build upon this study in several ways. One important direction is the use of larger and more diverse datasets, which would improve model generalization and provide a more robust evaluation of performance. Incorporating additional data sources, such as publicly available cheminformatics databases, could enhance the representativeness of the dataset and allow for more reliable conclusions. Another potential improvement is the exploration of alternative machine learning methods, such as Support Vector Machines, Gradient Boosting, or deep learning approaches, which may capture more complex relationships and further improve predictive performance.

In addition, feature engineering techniques could be applied to reduce dimensionality and improve model efficiency, particularly for structural fingerprints. Methods such as feature selection or dimensionality reduction could help identify the most informative descriptors and reduce noise. Further analysis of feature importance, especially using Random Forest, could

provide valuable chemical insights into which structural fragments or physicochemical properties are most strongly associated with antihistamine activity.

Finally, the methodology developed in this project can be extended to other drug discovery problems. Applying the same approach to different classes of compounds would test the transferability of the models and validate whether the observed patterns generalize beyond antihistamines. This would further strengthen the practical relevance of the work and demonstrate its applicability in broader pharmaceutical contexts.

In conclusion, this study confirms that Random Forest is the most effective model for predicting antihistamine activity within the scope of this dataset, and that combining structural and physicochemical descriptors leads to improved performance. The findings highlight the importance of both model selection and feature representation, and provide a solid foundation for future research in machine learning-based drug activity prediction.

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Appendix

Appendix A: Statistical Significance Analysis

This section contains the full WEKA Experimenter outputs used for statistical comparison of models. Symbols used in the analysis: * indicates that a model is significantly better than the baseline, while v indicates that a model is significantly worse than the baseline

A1. Base model random forest, comparison using percent correct

Random Forest is the reference model here. J48 is significantly worse on the merged dataset (v) but significantly better on Dataset 1 (*), while Logistic Regression shows no significant difference from Random Forest on any dataset.

Dataset	(2) trees.RandomFo	(1) trees.J48	(3) functions.L
merged_dataset	(100) 84.39 (6.33)	100.00 (0.00) v	83.38 (7.06)
dataset1_clean	(100) 80.66 (7.22)	72.20 (7.08) *	79.12 (7.91)
dataset2_clean	(100) 74.30 (7.67)	72.90 (7.00)	69.57 (7.68)
	(v/ /*)	(1/1/1)	(0/3/0)

A2. Base model random forest, comparison using F-measure

Using Random Forest as the baseline, J48 is significantly worse on the merged dataset (v), but significantly better on Dataset 1 (*). Logistic Regression is significantly better than Random Forest on Dataset 2 (*) and not significantly different on the other two datasets.

Dataset	(2) trees.RandomF	(1) trees.J48	(3) functions.
merged_dataset	(100) 0.84 (0.07)	1.00 (0.00) v	0.83 (0.08)
dataset1_clean	(100) 0.81 (0.08)	0.72 (0.07) *	0.79 (0.08)
dataset2_clean	(100) 0.74 (0.08)	0.72 (0.08)	0.66 (0.10) *
	(v/ /*)	(1/1/1)	(0/2/1)

A3. Base model random forest, comparison using ROC

With Random Forest as the base, J48 is significantly worse on the merged dataset (v) but significantly better on Dataset 1 and Dataset 2 (*). Logistic Regression is significantly better than Random Forest on Dataset 2 (*) and not significantly different on the merged dataset or Dataset 1.

Dataset	(2) trees.RandomF		(1) trees.J48	(3) functions.	
merged_dataset	(100)	0.92 (0.05)		1.00 (0.00) v	0.91 (0.05)
dataset1_clean	(100)	0.87 (0.07)		0.73 (0.09) *	0.86 (0.07)
dataset2_clean	(100)	0.82 (0.07)		0.73 (0.09) *	0.73 (0.09) *
		(v/ /*)		(1/0/2)	(0/2/1)

A4. Base model J48, comparison using percent correct

J48 is the baseline in this result. Random Forest and Logistic Regression are significantly worse on the merged dataset (* under the J48-based table), but significantly better than J48 on Dataset 1 (v), while no significant differences are found on Dataset 2.

Dataset	(1) trees.J48 '-C 0		(2) trees.Rando	(3) functions.L	
merged_dataset	(100)	100.00 (0.00)		84.39 (6.33) *	83.38 (7.06) *
dataset1_clean	(100)	72.20 (7.08)		80.66 (7.22) v	79.12 (7.91) v
dataset2_clean	(100)	72.90 (7.00)		74.30 (7.67)	69.57 (7.68)
		(v/ /*)		(1/1/1)	(1/1/1)

A5. Base model J48, comparison using F-measure

Relative to J48, both Random Forest and Logistic Regression are significantly worse on the merged dataset (*), but both significantly outperform J48 on Dataset 1 (v). No significant difference appears on Dataset 2.

Dataset	(1) trees.J48 '-C		(2) trees.Rand	(3) functions.	
merged_dataset	(100)	1.00 (0.00)		0.84 (0.07) *	0.83 (0.08) *
dataset1_clean	(100)	0.72 (0.07)		0.81 (0.08) v	0.79 (0.08) v
dataset2_clean	(100)	0.72 (0.08)		0.74 (0.08)	0.66 (0.10)
		(v/ /*)		(1/1/1)	(1/1/1)

A6. Base model J48, comparison using ROC

With J48 as the reference, Random Forest and Logistic Regression are significantly worse on the merged dataset (*), while both significantly outperform J48 on Dataset 1 (v). On Dataset 2, Random Forest is significantly better than J48 (v), whereas Logistic Regression is not significantly different.

Dataset	(1) trees.J48 '-C (2) trees.Rand (3) functions.		
merged_dataset	(100)	1.00 (0.00) 0.92 (0.05) *	0.91 (0.05) *
dataset1_clean	(100)	0.73 (0.09) 0.87 (0.07) v	0.86 (0.07) v
dataset2_clean	(100)	0.73 (0.09) 0.82 (0.07) v	0.73 (0.09)
		(v/ /*)	(2/0/1) (1/1/1)

A7. Base model logistic, comparison using percent correct

Logistic Regression is the baseline here. J48 is significantly worse on the merged dataset (v) but significantly better on Dataset 1 (*), while Random Forest does not differ significantly from Logistic Regression on any dataset.

Dataset	(3) functions.Logi (1) trees.J48 '- (2) trees.Rando		
merged_dataset	(100)	83.38 (7.06) 100.00 (0.00) v	84.39 (6.33)
dataset1_clean	(100)	79.12 (7.91) 72.20 (7.08) *	80.66 (7.22)
dataset2_clean	(100)	69.57 (7.68) 72.90 (7.00)	74.30 (7.67)
		(v/ /*)	(1/1/1) (0/3/0)

A8. Base model logistic, comparison using F-measure

Compared with Logistic Regression, J48 is significantly worse on the merged dataset (v) but significantly better on Dataset 1 (*). Random Forest is significantly better than Logistic Regression on Dataset 2 (v), while the two models are not significantly different on the merged dataset or Dataset 1.

Dataset	(3) functions.Log (1) trees.J48 (2) trees.Rand		
merged_dataset	(100)	0.83 (0.08) 1.00 (0.00) v	0.84 (0.07)
dataset1_clean	(100)	0.79 (0.08) 0.72 (0.07) *	0.81 (0.08)
dataset2_clean	(100)	0.66 (0.10) 0.72 (0.08)	0.74 (0.08) v
		(v/ /*)	(1/1/1) (1/2/0)

A9. Base model logistic, comparison using ROC

Using Logistic Regression as the base, J48 is significantly worse on the merged dataset (v) but significantly better on Dataset 1 (*). Random Forest is significantly better than Logistic Regression on Dataset 2 (v), while there is no significant difference between them on the merged dataset or Dataset 1.

Dataset	(3) functions.Log		(1) trees.J48	(2) trees.Rand
merged_dataset	(100)	0.91 (0.05)	1.00 (0.00) ▽	0.92 (0.05)
dataset1_clean	(100)	0.86 (0.07)	0.73 (0.09) *	0.87 (0.07)
dataset2_clean	(100)	0.73 (0.09)	0.73 (0.09)	0.82 (0.07) ▽
		(▽/ /*)	(1/1/1)	(1/2/0)

Appendix B: WEKA Knowledge Flow for Random Forest classification and evaluation

The figure illustrates the workflow used in WEKA's Knowledge Flow environment for model training and evaluation. The process begins with loading the dataset using the ARFF Loader, followed by class assignment and filtering. The data is then passed to a Cross-Validation Fold Maker. These are fed into the Random Forest classifier for model training and prediction. The output is evaluated using the Performance Evaluator, which generates both textual results and visual outputs. The Model Performance Chart provides graphical evaluation such as ROC curves, while the Text Viewer displays detailed classification metrics.

